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Experiment No.	5 (Home Work)

AIM:	SMTP Server Configuration
Part 1	
PROBLEM STATEMENT :	1. Two SMTP server programs, comp.com and cse.com. 2. Each Server has 2 users identified as ● a. user1@comp.com ● b. user2@comp.com ● c. user3@cse.com ● d. user4@cse.com 3. Each user should be able to send mail to any of the remining 3 users. 4. The mailbox should be maintained for each user by respective servers. And it should store mail. (Inbox and Sent) 5. The terminal will act as your User Agent for Accessing and Composing Mails. Scenario 1: Same Machine Both Server and their associated Users are running on the same PC. Scenario 2: Different Machine Both Server and their associated Users are running on different PC on same LAN.
Theory :	SMTP (Simple Mail Transfer Protocol) is an application-layer protocol used for sending electronic mail over a network. It follows a client–server architecture in which a mail client (user agent) sends messages to a mail server, and the server forwards them to the recipient's mail server. In this experiment, SMTP functionality is simulated using TCP socket programming in C instead of real mail servers. Two independent SMTP servers are created: ● **comp.com server** ● **cse.com server** Each server maintains two users and their mailboxes (Inbox and Sent). Users can send messages to any other user, whether on the same server or another

	server. The terminal acts as a Mail User Agent (MUA), allowing users to: ● compose mail ● send mail ● read inbox ● view sent mail Mail transfer is performed using TCP sockets: ● Client socket → sends mail request ● Server socket → receives and stores mail ● Inter-server communication → forwards mail between domains Thus the program demonstrates the basic working of SMTP: 1. User composes mail 2. Client connects to sender's SMTP server 3. Server checks recipient domain 4. If local → deliver to mailbox If remote → forward to other SMTP server 5. Mail stored in recipient Inbox and sender Sent Two execution scenarios are supported: **Same Machine:** Both SMTP servers and users run on one computer using different ports. **Different Machine (LAN):** Each SMTP server runs on a separate computer, and communication occurs over LAN using IP addresses. This experiment illustrates how email systems manage users, domains, mail transfer, and mailbox storage using socket-based communication in C.
PROGRAM:	Comp server.c

```c
#include <stdio.h>
#include <winsock2.h>
#include <string.h>

#pragma comment(lib, "ws2_32.lib")

#define PORT 2525

void save_mail(char *user, char *msg, char *type)
{
```

	(code below)

```
char path[100];
sprintf(path, "comp/%s_%s.txt", user, type);

FILE *fp = fopen(path, "a");
fprintf(fp, "%s\n------------------\n", msg); fclose(fp);
}

int main()
{
WSADATA wsa;
SOCKET server, client; struct
    sockaddr_in addr; char
    buffer[1024];

WSAStartup(MAKEWORD(2,2), &wsa);
server = socket(AF_INET, SOCK_STREAM, 0); addr.sin_family
    = AF_INET;
addr.sin_port = htons(PORT);
    addr.sin_addr.s_addr = INADDR_ANY;

bind(server, (struct sockaddr*)&addr, sizeof(addr));
    listen(server, 5);

    printf("comp.com SMTP Server running on port %d...\n",
PORT);

while(1)
    {
client = accept(server, NULL, NULL);

recv(client, buffer, sizeof(buffer), 0); printf("Mail
        received:\n%s\n", buffer);

// Example: TO:user1 char
        user[20];
```

```
sscanf(buffer, "TO:%s", user); save_mail(user,

        buffer, "inbox");

send(client, "250 OK", 6, 0);
        closesocket(client);
    }
}
```

User_agent.c

```
#include <stdio.h>
#include <winsock2.h>
#include <string.h>

#pragma comment(lib, "ws2_32.lib")

void send_mail(char *server_ip, int port, char *from, char
*to, char *msg)
{
WSADATA wsa; SOCKET
    s;
struct sockaddr_in server; char
    buffer[1024];

WSAStartup(MAKEWORD(2,2), &wsa);

s = socket(AF_INET, SOCK_STREAM, 0);

server.sin_family = AF_INET; server.sin_port =
    htons(port);
server.sin_addr.s_addr = inet_addr(server_ip); connect(s,

    (struct sockaddr*)&server, sizeof(server));

    sprintf(buffer, "FROM:%s\nTO:%s\nMSG:%s", from, to,
msg);
```

```
send(s, buffer, strlen(buffer), 0);

recv(s, buffer, sizeof(buffer), 0);
    printf("Server: %s\n", buffer);

closesocket(s);
}

int main()
{
int choice;
char from[50], to[50], msg[200];


printf("1.user1@comp.com\n2.user2@comp.com\n3.user3@cse.co
m\n4.user4@cse.com\n");
printf("Select sender: ");
    scanf("%d", &choice);

if(choice==1)     strcpy(from,"user1");
    if(choice==2) strcpy(from,"user2");
    if(choice==3) strcpy(from,"user3");
    if(choice==4) strcpy(from,"user4");

printf("Enter recipient (user1/user2/user3/user4): ");
    scanf("%s", to);

printf("Enter message: ");
    scanf(" %[^\n]s", msg);

// Domain routing
    if(strcmp(to,"user1")==0 || strcmp(to,"user2")==0)
        send_mail("127.0.0.1",2525,from,to,msg);
else
```

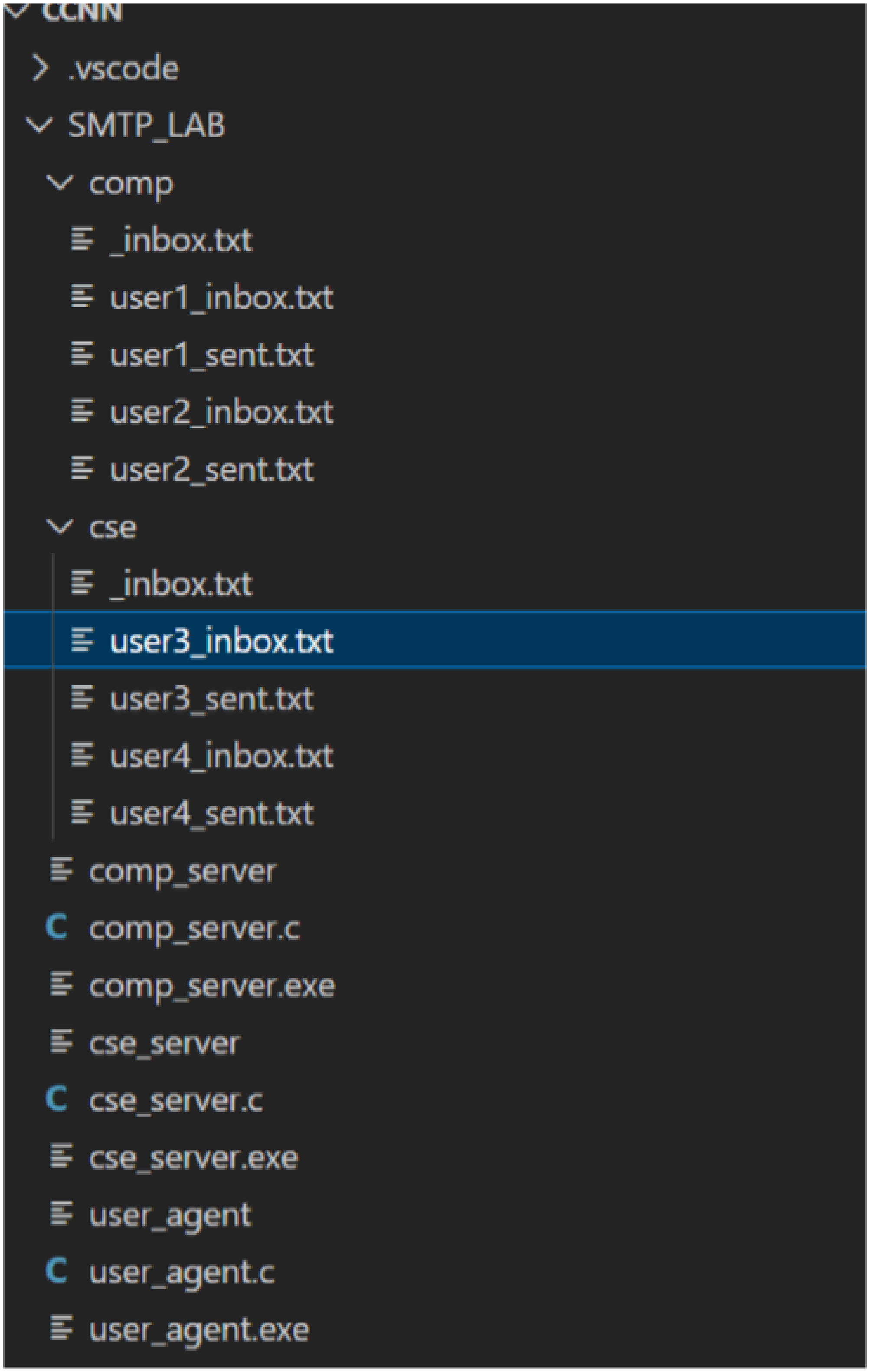
Cse_server.c

```
#include <stdio.h>
#include <winsock2.h>
#include <string.h>

#pragma comment(lib, "ws2_32.lib")

#define PORT 3535

void save_mail(char *user, char *msg, char *type)
{
char path[100];
sprintf(path, "cse/%s_%s.txt", user, type);

FILE *fp = fopen(path, "a");
fprintf(fp, "%s\n------------------\n", msg); fclose(fp);
}

int main()
{
WSADATA wsa;
SOCKET server, client; struct
    sockaddr_in addr; char
    buffer[1024];

WSAStartup(MAKEWORD(2,2), &wsa);
server = socket(AF_INET, SOCK_STREAM, 0); addr.sin_family
    = AF_INET;
addr.sin_port = htons(PORT);
    addr.sin_addr.s_addr = INADDR_ANY;

bind(server, (struct sockaddr*)&addr, sizeof(addr));
    listen(server, 5);
```

```
    printf("cse.com SMTP Server running on port %d...\n",
PORT);

    while(1)
    {
        client = accept(server, NULL, NULL);

        recv(client, buffer, sizeof(buffer), 0);
        printf("Mail received:\n%s\n", buffer);

        char user[20];
        sscanf(buffer, "TO:%s", user);

        save_mail(user, buffer, "inbox");

        send(client, "250 OK", 6, 0);
        closesocket(client);
    }
}
```

RESULT:

File Structure :


CCNN
.vscode
SMTP_LAB
comp
_inbox.txt
user1_inbox.txt
user1_sent.txt
user2_inbox.txt
user2_sent.txt
cse
_inbox.txt
user3_inbox.txt
user3_sent.txt
user4_inbox.txt
user4_sent.txt
comp_server
comp_server.c
comp_server.exe
cse_server
cse_server.c
cse_server.exe
user_agent
user_agent.c
user_agent.exe

Comp run :

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\DS\ccnn>cd SMTP_LAB

C:\DS\ccnn\SMTP_LAB>comp_server.exe
comp.com SMTP Server running on port 2525...
Mail received:
FROM:user2
TO:user2
MSG:Hellloooo
Mail received:
FROM:user4
TO:user1
MSG:Hellllllooooo
```

Cse Run :

```
C:\DS\ccnn\SMTP_LAB>cse_server.exe
cse.com SMTP Server running on port 3535...
Mail received:
FROM:user1
TO:user
MSG:2
Mail received:
FROM:user1
TO:3
MSG:Hellooooooo
Mail received:
FROM:user2
TO:user4@cse.com
MSG:hello
Mail received:
FROM:user1
TO:user4
MSG:TestMSG:hello
```

User run :

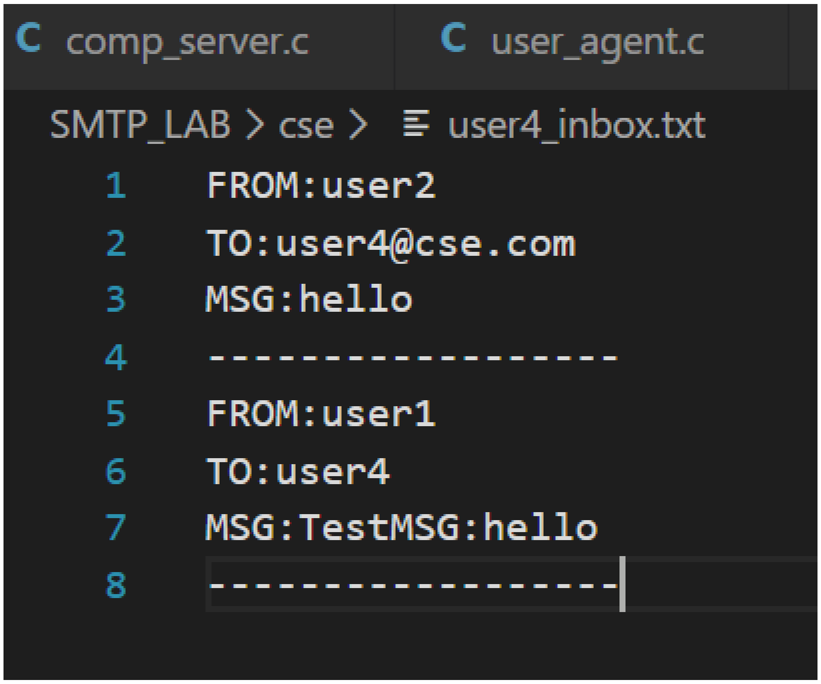
```
C:\DS\ccnn\SMTP_LAB>user_agent.exe
1.user1@comp.com
2.user2@comp.com
3.user3@cse.com
4.user4@cse.com
Select sender: 2
Enter recipient (user1/user2/user3/user4): user2
Enter message: Hellloooo
Server: 250 OKser2
TO:user2
MSG:Hellloooo

C:\DS\ccnn\SMTP_LAB>user_agent.exe
1.user1@comp.com
2.user2@comp.com
3.user3@cse.com
4.user4@cse.com
Select sender: 2
Enter recipient (user1/user2/user3/user4): user4@cse.com
Enter message: hello
Server: 250 OKser2
TO:user4@cse.com
MSG:hello

C:\DS\ccnn\SMTP_LAB>user_agent.exe
1.user1@comp.com
2.user2@comp.com
3.user3@cse.com
4.user4@cse.com
Select sender: 1
Enter recipient (user1/user2/user3/user4): user4
Enter message: Test
Server: 250 OKser1
```

OUTPUT Txt :

```
comp_server.c   user_agent.c
C:\DS\ccnn\SMTP_LAB\comp_server.c .txt
FROM:user4
TO:user1
MSG:Helllllooooo
------------------
```

```
comp_server.c   user_agent.c   cse_serve
SMTP_LAB > comp > user2_inbox.txt
FROM:user1
TO:user
MSG:2
------------------
FROM:user2
TO:user2
MSG:Hellloooo
------------------
```

```
comp_server.c   user_agent.c
SMTP_LAB > cse > user3_inbox.txt
FROM:user1
TO:3
MSG:Helloooooo
------------------
```



Part 2

PROBLEM STATEMENT :	**Home Assignment** 1. Use Sendmail/Postfix package in Linux to configure SMTP server. 2. A mail must be received on your SPIT email from your personal Gmail account, using your configured SMTP Server.
Theory :	**Simple Mail Transfer Protocol (SMTP)** SMTP (Simple Mail Transfer Protocol) is an application-layer protocol used for sending email over TCP/IP networks. It operates on port 25 (default), port 587 (submission), or port 465 (secure SMTP). SMTP follows a client–server model: ● **Mail User Agent (MUA)** → User email client (Gmail, Outlook, etc.) ● **Mail Transfer Agent (MTA)** → Mail server (Postfix, Sendmail) ● **Mail Delivery Agent (MDA)** → Delivers mail to mailbox Mail flow: Sender → SMTP Server → Internet → Receiver Mail Server → Recipient Inbox **Postfix Mail Server** Postfix is a Mail Transfer Agent (MTA) used in Linux systems to send and receive

emails.
It is widely used because it is secure, fast, and easy to configure.
Main configuration file:
/etc/postfix/main.cf
Postfix roles:

- Accepts mail from clients
- Routes mail to destination server
- Delivers mail locally or remotely

Procedure / Steps
Step 1: Install Postfix
Step 2: Start and Enable Postfix
Step 3: Verify SMTP Port
Step 4: Send Test Mail from Linux SMTP Server
Step 5: Send Mail from Gmail via Configured SMTP

1. Open Gmail
2. Click **Compose**
3. Send mail to:

yourSPITemail@spit.ac.in

4. Subject:

SMTP Server
Test

5. Message:

Step 6: Verify Mail Delivery
Login to SPIT email → Inbox

RESULT:

Installing Postfix Mailutils :

```
sudo apt install postfix mailutils
[sudo] password for student:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Check Postfix Status :

```
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (exited) since Mon 2026-03-02 15:36:33 UTC; 29s ago
       Docs: man:postfix(1)
    Process: 1668 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1668 (code=exited, status=0/SUCCESS)
        CPU: 3ms

Mar 02 15:36:33 Shantanu systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Mar 02 15:36:33 Shantanu systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
student@Shantanu:/mnt/c/DS/ccnn/SMTP_LAB$
```

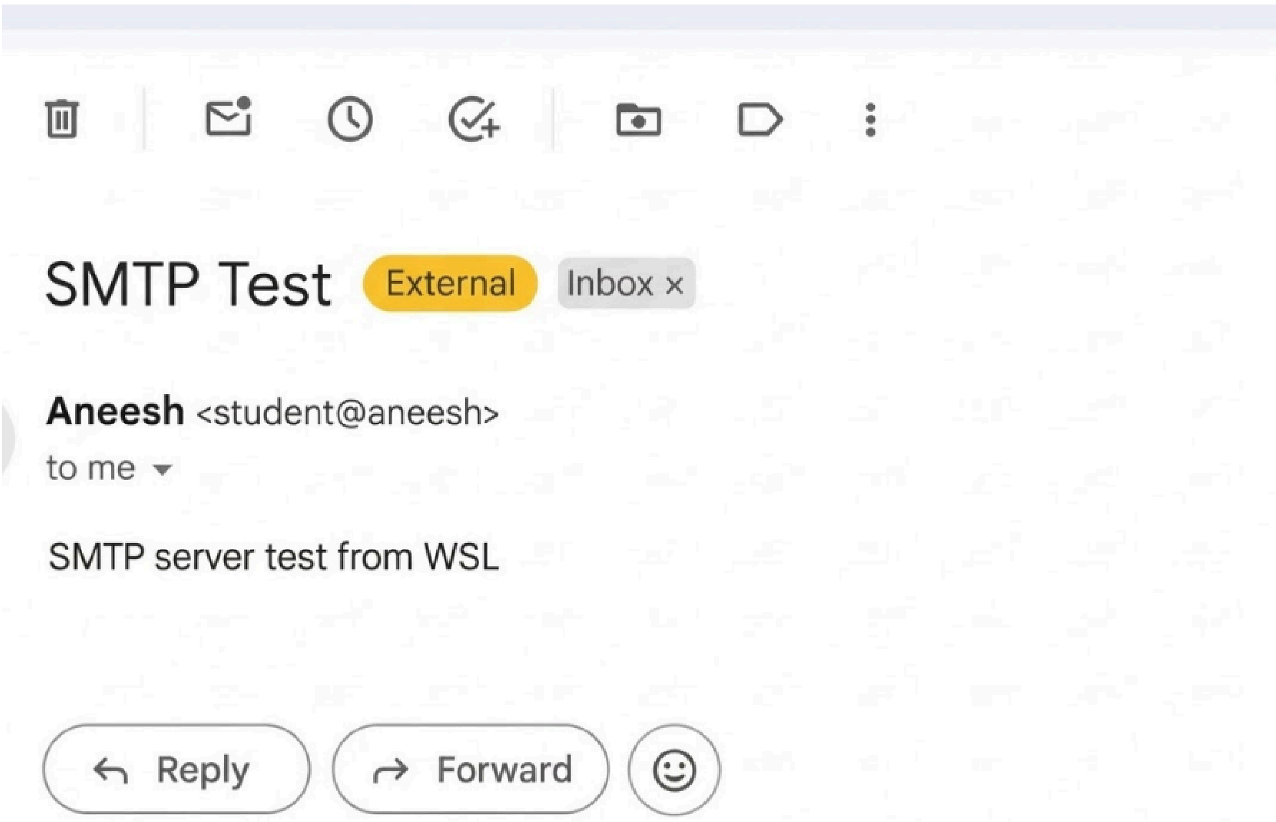
Check Port :

```
LISTEN 0      100          0.0.0.0:25        0.0.0.0:*    users:(("master",pid=1664,fd=13))
LISTEN 0      100             [::]:25           [::]:*    users:(("master",pid=1664,fd=14))
student@Shantanu:/mnt/c/DS/ccnn/SMTP_LAB$
```

Sending Email :

```
echo "SMTP server test from WSL" | mail -s "SMTP Test" aneesh.suvarna24@spit.ac.in
```

Final Gmail OUTPUT :



CONCLUSION:	This experiment demonstrated the working of SMTP using TCP socket programming in C by creating two mail servers and multiple users. It showed how emails can be composed, transmitted between domains, and stored in user mailboxes (Inbox and Sent). The implementation helped in understanding client–server communication and basic mail transfer mechanisms. Thus, the objectives of simulating an SMTP system were successfully achieved.